

Classics in Chemical Neuroscience: Amitriptyline

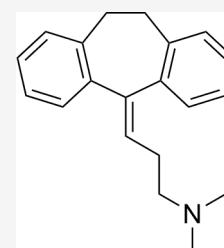
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ABSTRACT: Amitriptyline was the second tricyclic antidepressant to appear on the market for major depressive disorder under the brand name Elavil in 1961. Since its emergence, amitriptyline has been an effective therapeutic in various disease states and disorders but has also been a concerning source of cardiotoxicity. Amitriptyline inhibits serotonin and norepinephrine reuptake as well as produces off-target activity at histaminergic, muscarinic, and various other receptors. Its role as a modulator of monoamines helped further establish the monoamine theory to understand various mood disorders, paving the way for the now more common selective serotonin/norepinephrine reuptake inhibitors. In this review, we will discuss amitriptyline's synthesis, manufacturing information, drug metabolism, pharmacology, adverse effects, and its history and importance in therapy to present amitriptyline as a true classic in chemical neuroscience.



Amitriptyline

KEYWORDS: Amitriptyline, major depressive disorder, depression, tricyclic antidepressant, Elavil, nortriptyline

I. INTRODUCTION

Major depressive disorder (MDD) is a psychiatric condition that affects mood and cognition. It presents mainly as increased feelings of negativity and a generalized decrease in satisfaction from previously enjoyed activities that causes significant impairment in daily life.^{1–3} With severity ranging from mild to severe, MDD's diagnosis is based on a collection of various symptoms that occur for at least 2 weeks. These include lethargy, appetite changes, suicidal ideation, among many other symptoms deleterious to one's mood.^{1,2} MDD occurs in approximately 1 in 5 adults throughout their lifetime [onset time, mean (SE) = 29 years (0.21)]. Females are more likely to be diagnosed with MDD than males, and approximately half of those diagnosed with MDD also present with substance-abuse disorders. Comorbidity with anxiety disorders or personality disorders occurs in roughly a third of those affected by MDD.^{4–8}

In 1959, tricyclic antidepressants (TCAs) emerged with imipramine (1, Figure 1), which provided a new pharmaceutical class for treating MDD.⁹ The following year, imipramine's diamine structure was substituted with an alkene at the central

amine to create amitriptyline (2). Amitriptyline was approved for the treatment of MDD in 1961 by the United States Food and Drug Administration (FDA) under the brand name Elavil.¹⁰

Amitriptyline is included in the World Health Organization's List of Essential Medicines of 2019, providing a broad medicinal profile with a long list of off-label indications, despite only being FDA-approved for MDD.^{11,12}

Current treatments vary (cognitive behavioral therapy, pharmaceuticals, classical psychedelics, etc.), but it was not until the latter half of the 20th century that some of the pharmacological underpinnings of MDD became more apparent.^{13,14} The historical significance of antidepressants, specifically amitriptyline's impact, will be explored more thoroughly in [History and Importance in Neuroscience](#).

In this review, we will organize available information on amitriptyline's synthesis, drug metabolism, pharmacology, and importance for neuroscience. We also will present data from manufacturing and sales. Whereas there are other resources^{15–19} for an overview of amitriptyline, the goal of this review is to compile relevant, up-to-date data into one easily accessible format in order to demonstrate amitriptyline's presence as a classic of chemical neuroscience.

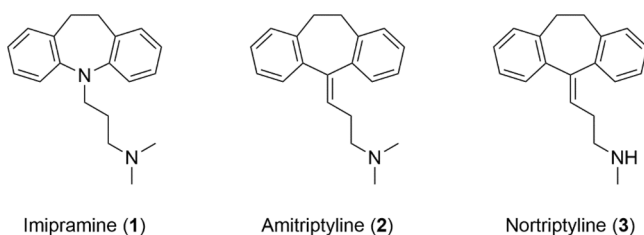
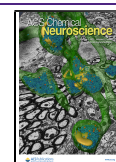


Figure 1. Commonly prescribed tricyclic antidepressants.

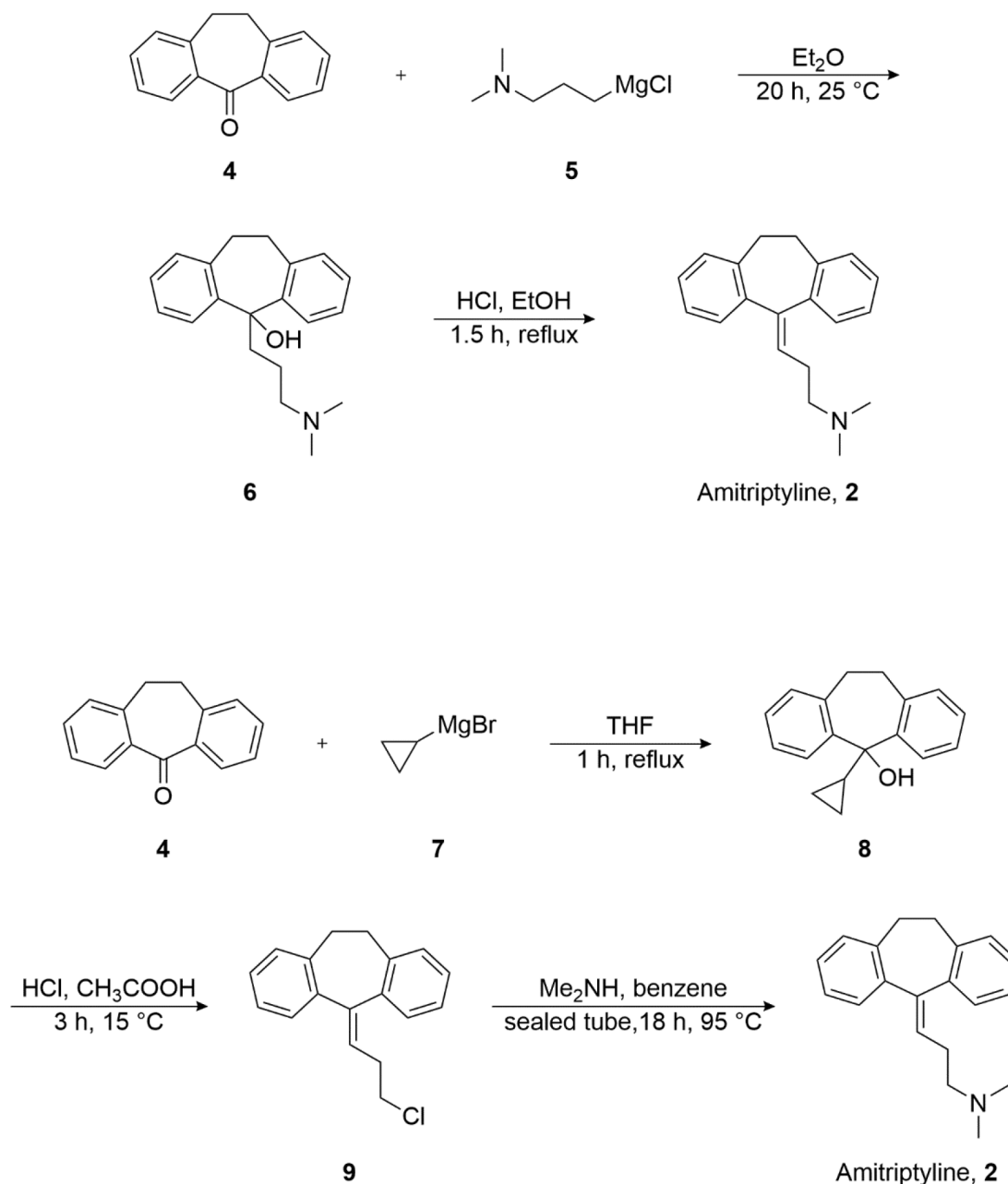
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Scheme 1. Chemical Syntheses of Amitriptyline, 2



II. CHEMICAL SYNTHESIS

Amitriptyline (CAS no. 50-48-6) [3-(10,11-dihydro-5H-dibenzo[*a,d*][7]annulen-5-ylidene)-*N,N*-dimethylpropan-1-amine] [$\text{C}_{20}\text{H}_{23}\text{N}$] is a small (MW = 277.41 Da) tertiary amine tricyclic antidepressant with no hydrogen bond donors and a single hydrogen bond acceptor.¹⁹

The original patent literature shows two main routes for the synthesis of amitriptyline (2) starting from dibenzosuberone (4) as shown in Scheme 1. The first synthesis of 2 commences with 4 undergoing a reaction with the Grignard reagent (5) derived from dimethylaminopropyl chloride to yield the tertiary alcohol (6).²⁰ Compound 6 is then dehydrated to amitriptyline (2) using acidic conditions.²¹

An alternative synthesis was also reported in which 4 undergoes a reaction with the Grignard reagent (7) derived from cyclopropyl bromide to yield the tertiary alcohol (8).²²

Compound 8 then quantitatively rearranges to the primary chloride (9) upon treatment with a solution of hydrogen chloride in acetic acid. Ultimately, 2 is obtained from 9 by reaction with dimethylamine in a sealed tube.

III. MANUFACTURING INFORMATION

Amitriptyline is currently only available to the public as a generic hydrochloride salt; brand names have included Elavil (originally manufactured by AstraZeneca), Endep (manufactured by Roche), Amitril (manufactured by WL/PD), and Saroten (manufactured by Warner).²³ The FDA discontinued each of these amitriptyline brand names (and many others containing amitriptyline) in the early 2000s due to cardiotoxic concerns,²⁴ although in 2017 the agency issued a notice stating that the Elavil tablets were discontinued for reasons not previously stated. Under pressure from a citizen petition filed

by Alembic Pharmaceuticals Limited, the FDA reviewed agency records and concluded that “the petitioner has identified no data or other information suggesting that these products were withdrawn for reasons of safety or effectiveness”.²⁵

Pharmaceutical manufacturers now are able to apply for abbreviated new drug applications with formulations containing amitriptyline hydrochloride.²⁵ However, cardiotoxicity remains a well-established concern as discussed in *Pharmacology, Adverse Effects, and Dosages*.

Amitriptyline can also be prescribed as a combination therapy tablet with chlorthalidone (a benzothiazide) or perphenazine (an antipsychotic), both produced by Mylan Pharmaceuticals Inc.²³ It currently is commercially available as a tablet in combination therapy as a hydrochloride salt to aid in its absorption, but when injected intramuscularly in a hospital setting, the compound is usually presented as an embonate (pamoate) salt.²⁶

In 2016 amitriptyline was the 88th most prescribed medication in the United States, with over 8.6 million prescriptions and total patient out-of-pocket costs exceeding \$60 million.²⁷ Average wholesale costs per unit for generic amitriptyline hydrochloride sold in 100-tablet bottles are estimated in Table 1 as of November 2020.²⁸ Prescription

Table 1. Amitriptyline Unit Prices

tablet strength (mg)	average wholesale unit price (\$)
10	0.32
25	0.64
50	1.27
75	1.91
100	2.54
150	3.81

prices must be affordable in order to effectively treat a disorder that does not differentiate based on socioeconomic status. Amitriptyline's price typically does not serve as a barrier for treatment.

IV. DRUG METABOLISM

Amitriptyline is readily absorbed in the GI tract, resulting in an initial bioavailability of 90–95%, decreasing to a mean bioavailability of 53% after first pass metabolism ($F = 0.527 \pm 0.123$).^{11,29} The main hepatic metabolizing enzymes of clinical significance for amitriptyline are CYP2C19 and CYP2D6. CYP2C19 is the primary enzyme responsible for demethylation of amitriptyline into its main metabolite nortriptyline (3), which has a similar pharmacological profile, and then 3 is demethylated to yield demethylnortriptyline (10, Figure 2). CYP2D6 contributes to the hydroxylation of nortriptyline (3) to its most common metabolites, (Z)-10-hydroxynortriptyline (11) and (E)-10-hydroxynortriptyline (12), with the (E)-enantiomer produced at approximately 5 times the amount of the (Z)-enantiomer. The hydroxynortriptyline metabolites are nontherapeutic and have been associated with amitriptyline's cardiotoxicity.^{30,31}

Though CYP2D6 and CYP2C19 are the most common P450 enzymes related to amitriptyline's metabolism, polymorphisms of these enzymes are not predictive of any one specific adverse event. Pharmacogenomic testing can help to explain differences in amitriptyline's efficacy among MDD patients, however.³¹ It is recommended to avoid prescribing

amitriptyline for patients who are CYP2D6 ultrarapid or poor metabolizers due to decreased efficacy or a general increase in toxic side effects, respectively.³²

Of clinical importance, monitoring plasma levels of the hydroxy metabolites of both amitriptyline and nortriptyline has predictive value in clinical outcomes of inpatients suffering from MDD. An increase in 11 and (Z)-10-hydroxyamitriptyline (13) plasma concentrations helps predict positive psychiatric outcomes based on a global assessment of functioning (GAF) test utilizing DSM-III clinical outcomes from 1997. Additionally, an increase in 12 and (E)-10-hydroxyamitriptyline (14) metabolites is associated with negative outcomes based on the same measures.³³ However, because of updated diagnostic and clinical presentations within the DSM-V, these findings may not be entirely sound within a modern psychological perspective.

Even still, pharmacogenetic testing of MDD patients could be used to identify those that could gain the most benefit from this medication and/or those more prone to adverse events from amitriptyline. This would not only decrease suffering for the patient but also decrease the likelihood of unnecessary prescriptions.^{34,35}

V. PHARMACOLOGY, ADVERSE EFFECTS, AND DOSAGES

Tricyclic antidepressants appear to relieve depressive symptoms primarily through regulation of monoamines, specifically serotonin and norepinephrine. Amitriptyline inhibits serotonin and norepinephrine reuptake, though its primary metabolite, nortriptyline, is a more potent inhibitor of norepinephrine reuptake than serotonin. Amitriptyline inhibits the serotonin reuptake transporter (SERT, $K_i = 3\text{--}4\text{ nM}$) and the norepinephrine reuptake transporter (NET, $K_i = 19\text{--}35\text{ nM}$) (Table 2). Amitriptyline also has a relatively high affinity for 5-HT_{2c} and α_1 ($K_i = 4.0$ and 4.4 nM , respectively). All metabolites of amitriptyline are less anticholinergic than amitriptyline itself, though the 10-hydroxy metabolites (11, 12, 13, 14) are equipotent to amitriptyline in inhibiting norepinephrine reuptake.³⁶ These neurotransmitters and respective receptors have been implicated in the symptomatology of MDD in many reviews.^{3,10,36–38}

Monoamine regulation does not present the whole therapeutic picture, however. Growing evidence suggests depression symptomatology is very polygenic. Neurotrophic dysfunction and neuroimmune dysfunction appear to be prominent possible causes as well.^{42,43} Brain-derived neurotrophic factor (BDNF) is a key player for the neurotrophic hypothesis, promoting synaptogenesis and general neurodevelopment.^{43,44}

Many antidepressants have been found to increase BDNF within certain areas of the brain; amitriptyline increases BDNF mRNA expression but only in astrocytic and microglial cultures.^{43,45} Though amitriptyline's effect on BDNF does not appear to act directly on neurons, glial cell dysfunction and atrophy can still be large contributing factors in MDD neuropathophysiology.⁴⁶ Amitriptyline may exert its therapeutic effects both through the neurotrophic and through the monoamine hypotheses for MDD.

Amitriptyline also is prescribed off label for patients experiencing neuropathic pain. This seems in part due to its common binding site with local anesthetics, acting as a blocker of sodium channels and disrupting the normal signaling cascade of action potentials.⁴⁷ More importantly, amitriptyline

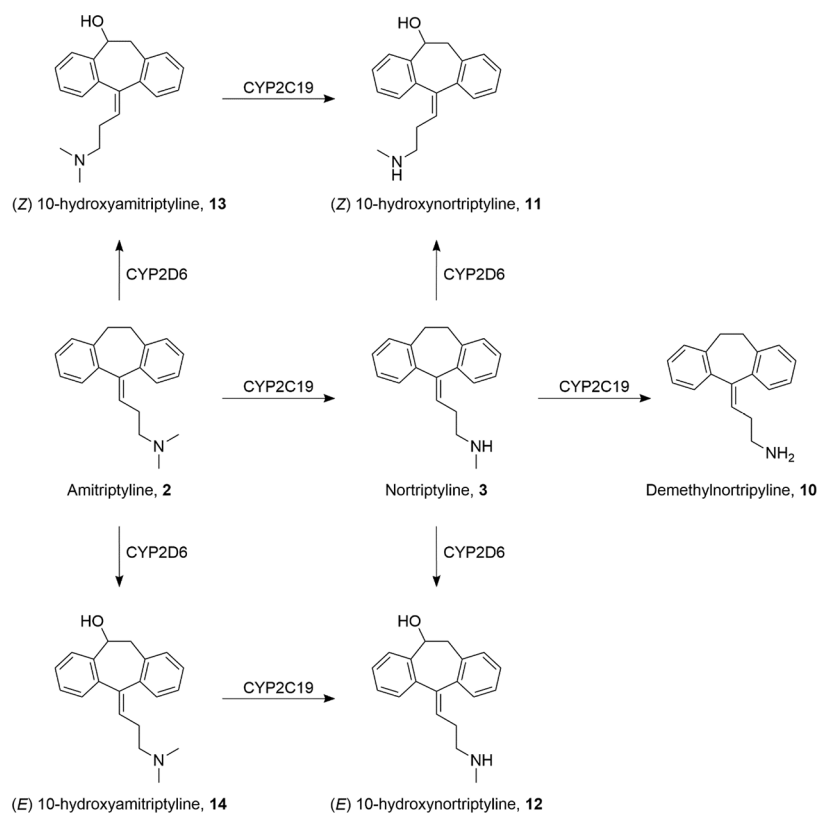


Figure 2. Primary phase I metabolite structures of amitriptyline, 2, and its major metabolite nortriptyline, 3.

Table 2. Binding Affinities of Amitriptyline and Nortriptyline on Common Targets^{36,39–41}

binding site	K_i (nM), amitriptyline	K_i (nM), nortriptyline
α_1	4.4	55
H ₁	0.5–1.1	3–15
M ₁	11–14.7	40
M ₂	11.8	110
M ₃	12.8–39	50
M ₄	7.2	84
M ₅	15.7–24	97
NET	19–35	1.8–4.4
SERT	2.8–4.3	15–18
σ_1	4.4–24	>2000
5-HT _{2a}	18–23	5.0–41
5-HT _{2c}	4.0	8.5

helps to increase the concentration of norepinephrine in the dorsal horn of the spinal cord. This norepinephrine reuptake inhibition increases the basal activity of α_2 receptors, which mediate an analgesic effect by increasing GABA transmission among spinal interneurons.⁴⁸ This effect from amitriptyline to treat acute pain from noxious stimuli is small. However, activation of α_2 receptors via other agents that increase norepinephrine in the spinal cord has shown pronounced effects in hyperalgesia associated with neuropathic pain.⁴⁹

Additional off-label indications include irritable bowel syndrome, insomnia, and acute psychosis in patients with schizophrenia. It is also the most widely researched medication for frequent tension headaches as well as a first-line treatment for fibromyalgia.^{50–55} Though commonly prescribed for headaches in children, a randomized, placebo-controlled study showed no significant reduction of headache frequency

or degree of disability associated with migraines in children and adolescents taking amitriptyline.⁵⁶

Compared to imipramine (1), amitriptyline contributes to double the unexpected death rate due to cardiotoxic complications.⁵⁷ It is not uncommon for amitriptyline to be taken in excess in an attempt to self-poison. Unfortunately, the same medication used to treat a patient's MDD also can be a method by which they could intentionally die by suicide.

Amitriptyline accounts for approximately 40% of antidepressant overdose fatalities, regardless of antidepressant class. It consistently has been in the top 20 drugs involved in overdose deaths in the United States.^{58,59} Patients presenting with tricyclic antidepressant overdose (confusion, arrhythmias, seizures) typically have taken a dose exceeding the minimum toxic dose (5 mg/kg for amitriptyline) but more commonly in excess of 20 mg/kg.⁵⁸ Therapeutic doses are much lower, as the maximum daily dose is 150 mg regardless of the patient's weight. For reference, that maximum daily dose for a 150 lb patient is 2.2 mg/kg.¹¹

Sudden cardiac death is not uncommon during TCA overdose. A frequent indicator of tricyclic antidepressant cardiotoxicity is an elongated QRS wave and QT interval. This is in part due to the (Z)-10-hydroxynortriptyline (11) metabolite, as it has been associated specifically with these two presentations of cardiotoxicity in the elderly.⁶⁰ This metabolite has also been linked with bradycardia and decreased cardiac output.⁶¹ The elongated intervals can be narrowed by an intravenous solution of sodium bicarbonate, and if the patient experiences a seizure, benzodiazepines are recommended.⁶²

The pathophysiology of amitriptyline's cardiotoxic effects is not fully understood. However, amitriptyline administration appears to increase sarcoplasmic reticulum release of calcium

via activation of ryanodine receptors in ventricular myocytes. This effect in combination with amitriptyline's ability to inhibit sarcoplasmic reuptake of calcium may contribute to the cardiotoxic effects commonly presented with tricyclic antidepressant overdose.^{63,64}

Additionally, the potassium channel encoded by the human ether-a-go-go-related gene (hERG) is a commonly screened binding site for drugs in preclinical development. Compounds that inactivate this channel are known to elongate the QT interval, a common presentation of amitriptyline overdose.^{60,65} Amitriptyline binds to the hERG potassium channel with an IC_{50} of $4.78 \pm 3.70 \mu M$ in mammalian physiological conditions.⁶⁶ It is unclear the extent amitriptyline's inhibition of the hERG potassium channel contributes to its mortality. Though arrhythmias are notable in amitriptyline overdoses, individuals who die after hospital admittance do so primarily because of direct suppression of myocardial function, not arrhythmias.⁶⁷

Less fatal, though serious, adverse effects strongly associated with amitriptyline include anticholinergic effects (dry mouth, slight dizziness), constipation, tachycardia, and confusion.^{68–70} These are most likely due to the compound's antagonism of adrenergic, muscarinic, and histaminergic receptors, contributing to the side effects of dizziness, memory impairments, and drowsiness, respectively.⁹

Taking amitriptyline also increases the likelihood of gaining weight. Patients can experience a heightened desire for carbohydrates as well as an increase in serum leptin levels, which can lead to weight gain by increasing caloric intake and promoting leptin resistance.^{71–73} While others have not seen the increase in serum leptin after amitriptyline administration, the sample size was much smaller in this study.⁷⁴

Amitriptyline is a potent antihistamine ($K_i = 0.5–1.1$ nM) and has a similar binding affinity for the H_1 receptor as many prescription antihistamines.^{75–77} This contributes to amitriptyline's sedation, but it also may contribute to weight gain. Histamine modulates the appetite-reducing effects of leptin. By blocking of H_1 receptors in the hypothalamus, leptin resistance develops more rapidly, in turn promoting basal appetite.⁷⁸ Antihistaminergic compounds have been shown to increase BMI, waist circumference, and weight.^{79,80}

The effect size of weight gain from amitriptyline is relatively moderate compared to other antidepressants in different classes on the market.^{81,82} However, it seems to promote weight gain more than other TCAs.⁸³ Ultimately, the mechanism by which amitriptyline increases weight is comprehensive and multifactorial.

Sexual dysfunction is estimated to occur in 6 times more males than females who take amitriptyline.⁸⁴ Sexuality is a complex and integrated part of one's identity. As such, emotional and psychological processes are as important as physical stimulation. Sexual dysfunction is mediated by serotonergic systems that interplay with dopaminergic and noradrenergic signaling in relation to sexual motivation and satisfaction, in both the peripheral and central nervous systems.^{85,86} Amitriptyline's central and peripheral effects on serotonin are the most likely culprits for inducing sexual dysfunction.

The initial dosage for outpatient adults with MDD is either 50 mg or 75 mg at their hour of sleep with increasing doses, if needed, until a maximum of 150 mg daily. Maintenance dosages are usually between 50 mg and 100 mg daily once the provider is aware of how the medication affects the patient.¹¹

Additionally, the combined plasma concentration of amitriptyline and nortriptyline significantly correlates to clinical improvement, as measured through the Hamilton depression rating scale.^{87,88} Therapeutic levels of amitriptyline range from 75 to 175 ng/mL, though commonly the combination of amitriptyline and its metabolite, nortriptyline, is measured with a broad therapeutic range of 60–220 ng/mL.^{89–91}

Individuals should not take amitriptyline within 2 weeks of discontinuing a monoamine oxidase-inhibitor (MAO-I) treatment, as a potentially lethal interaction through serotonin syndrome can occur. To decrease the likelihood of uncomfortable withdrawal effects, patients should avoid abruptly stopping amitriptyline. For this reason, it is considered by the FDA to be a class C pregnancy risk since neonates whose mothers took amitriptyline in the third trimester can experience withdrawal symptoms.¹¹ Additionally, the World Health Organization recommends that breastfeeding is compatible with the maximum accepted dose of 150 mg per day.⁹²

Amitriptyline should not be taken along with alcohol or other CNS depressants, since it can potentiate their effects. Additionally, those with a history of QT prolongation or other cardiovascular disorders should not consider amitriptyline as a treatment for MDD.^{10,11}

VI. HISTORY AND IMPORTANCE IN NEUROSCIENCE

MDD has been noted throughout history in various forms, from empirical literature to art. However, it was not until the 1900s that its scientific understanding began to flourish. The role of monoamines, specifically serotonin, in MDD became more clear in the 1960s when a lower concentration of serotonin in the hind-brain of depressed individuals was noticed in post-mortem analysis.⁹³ The monoamine hypothesis for depression helped set the groundwork for current gold-standard compounds that selectively increase the amount of serotonin and norepinephrine in synaptic clefts. However, the first class of antidepressants on the market was not developed out of this insight, as nonselective monoamine oxidase inhibitors were marketed prior to the understanding of monoamines' role in regulating mood.⁹

The first pharmacological treatment for MDD appeared in 1958 with the production of iproniazid, a MAO-inhibitor, marketed under the trade name Marsilid. This period was a very progressive time for psychopharmacology as TCAs emerged only a year later with the release of imipramine (1) under the brand name Tofranil.⁹ This drug's antidepressant effect was a serendipitous discovery. Chlorpromazine, structurally very similar to TCAs, was successfully treating patients with schizophrenia since its launch in 1952 in France. After structural explorations of chlorpromazine's pharmacophore, researchers experimented with imipramine to treat schizophrenia in the late 1950s. It was not effective at relieving psychotic symptoms, but a clinical psychiatrist noted in regard to his patients' moods that they "again [seek] contact with other people ... take part in games, become more cheerful, and are once again able to laugh..."⁹

Imipramine provided about the same amount of relief (but with less dangerous side-effects) as the MAOIs, thus the establishment of a new drug class for treating MDD. Amitriptyline's development was not a monumental chemical synthetic leap from imipramine. The replacement of a tertiary amine to a carbon was the extent of the modification. Within 2

years, amitriptyline was marketed by AstraZeneca for the treatment of MDD.¹⁰

TCAs are the only class of antidepressants that are not based on their mechanism of action, primarily due to a limited understanding of imipramine's pharmacodynamics at the time of release. More recent evidence presented in this review suggests that tricyclics have a broad pharmacological profile beyond their inhibition of amine reuptake transporters, with strong antihistaminergic and moderate anticholinergic activity. The disruption of amine reuptake transporters is likely what predominantly contributes to tricyclics' antidepressant effects.³⁷

Currently, selective monoamine reuptake inhibitors, such as selective serotonin reuptake inhibitors (SSRIs) and serotonin and norepinephrine reuptake inhibitors (SNRIs), are the first line of medications considered for MDD due to their efficacy and fewer side effects, as well as a decreased risk of overdose, compared to tricyclic and nonselective MAO-inhibitor alternatives.⁹⁴ However, MDD inpatients seem to benefit more from tricyclics, specifically amitriptyline, compared to SSRIs, because they can be administered 2–3 times the normal dose in a hospital setting due to the ability to monitor for serious adverse reactions.⁹⁵

Developing safe and effective antidepressants is relatively new. Amitriptyline was not the first, and it certainly will not be the last. However, it highlights a uniquely human motivation: the drive to maximize the well-being of others. This compound was a step along the medical progression in treating MDD, one of the most complex physiological disorders.

Amitriptyline can be an agent of extreme suffering also, not only for the patient but for the family of the individual who used it to facilitate taking their life. Complex challenges, such as MDD, require powerful and robust solutions. The ability of any compound to aid in recovery from MDD should be balanced with its potential to endanger the patient who ingests it. Nevertheless, pharmacotherapy for psychiatric disorders, such as MDD, schizophrenia, generalized anxiety disorders, and many others, continues to strive toward safer compounds that effectively relieve undue suffering from patients and those who love them.

As long as there are individuals who are treatment-resistant, there will be enough motivation to increase their well-being via new therapeutic mechanisms. Individuals who experience MDD present clinically in many different ways. Conservative estimates regarding the DSM-IV diagnostic criteria show 1497 different diagnostic combinations of an MDD episode.⁹⁶ This estimate does not consider severity, which can often alter the course of treatment. Clinicians need a wide range of possible therapeutics to treat patients with MDD because of the large variability between individuals. Thankfully, this motivates drug discovery teams to develop novel mechanistic models constantly.

New antidepressants are in the pipeline across many pharmaceutical entities, including NMDA antagonists (esketamine), metabotropic glutamate receptor negative allosteric modulators (mGluR NAMs), and selective inhibitors of MAO-A, a subtype of monoamine oxidase (moclobemide, bifemelane).⁹⁷ Additionally, classical psychedelics, such as psilocybin, LSD, and DMT, have reached various clinical milestones for MDD treatments.^{98–102} Though not novel compounds, psychedelics can be used in psychedelic assisted psychotherapy to offer an effective treatment paradigm, one focused on using one or two doses to dissolve the ego and

integrate the acute psychedelic experience via sober follow-up sessions with therapists.^{103–105}

Many exciting compounds and nonpharmaceutical therapies continue to increase in efficacy and accessibility. Amitriptyline's place in pharmaceutical history is still sound in the context of further developments, and for this reason, it is truly a classic in chemical neuroscience.

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E.W.M. researched all sections and wrote sections I, III, IV, V, and VI. R.N.D. researched and wrote section III and provided guidance and mentorship.

Notes

The authors declare no competing financial interest.

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